Physica A 544 (2020) 123453

Contents lists available at ScienceDirect

Physica A

journal homepage: www.elsevier.com/locate/physa

Synchronization of reservoir computers with applications to communications

Tongfeng Weng [a,*], Jia Song [a], Huijie Yang [a], Changgui Gu [a], Jie Zhang [b], Michael Small [c,d]

[a] Business School, University of Shanghai for Science and Technology, Shanghai 200093, China
[b] Centre for Computational Systems Biology, Fudan University, Shanghai 200093, China
[c] Complex Systems Group, Department of Mathematics and Statistics, The University of Western Australia, Crawley, Western Australia 6009, Australia
[d] Mineral Resources, CSIRO, Kensington, WA 6151, Australia

ARTICLE INFO

Article history:
Received 17 September 2019
Available online 6 November 2019

Keywords:
Synchronization
Reservoir computing approach
Chaotic system

ABSTRACT

We consider different reservoir computers learning and modeling the same chaotic system. We find that by transmitting a common signal, synchronization will be achieved between these superficially distinct reservoir computers. We further show that even when the transmitted signal is contaminated with extensive observational noise, synchronization will still occur. Moreover, utilizing a cascaded synchronization scheme, we can successfully recover a chaotic masking signal hidden in the transmitter. Our findings reveal that synchronization is a common feature of reservoir computers upon learning chaotic systems.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

It is widely agreed that machine learning technique is a critical branch of artificial intelligence. The importance originates from its powerful applications ranging from playing go [1] to facial recognition [2] and web searches [3]. A variety of machine learning techniques, such as support vector machine [4], decision trees [5], and deep learning [1], are emerging for dealing with the problems encounted in our life related to artificial intelligence. Among them, an intriguing one known as reservoir computing approach has recently attracted increasing attention [6–10]. In particular, intensive studies demonstrate that this approach has been used successfully for chaotic system prediction [7], inferring unmeasured variables [11], and attractor reconstruction [12]. In this sense, reservoir computing technique offers an alternative way for modeling and characterizing nonlinear dynamics, especially for chaotic systems.

When using reservoir computing technique for modeling a given chaotic time series, a number of reservoir parameters, such as a "leakage" rate, size of the reservoir network, and the number of the training data, need to be given in advance. However, different experts will typically construct not identical reservoir computers for capturing its evolutionary mechanism. In this context, an intriguing problem is what is the common feature linking these superficially distinct reservoir computers? In this paper, we demonstrate that synchronization is a common feature connecting reservoir computers upon learning the same chaotic system. Specifically, we show that by sending a common signal, the synchronization phenomenon will be present in the trained reservoir computers. In fact, synchronization is a universal concept in nonlinear sciences [13,14]. Moreover, we find that even when the transmitted signal is contaminated by extensive white or colored noise, synchronization will still occur. We further show that this synchronization feature has a potential application for chaotic masking.

* Corresponding author.
E-mail addresses: wtongfeng2006@163.com (T. Weng), michael.small@uwa.edu.au (M. Small).

https://doi.org/10.1016/j.physa.2019.123453
0378-4371/© 2019 Elsevier B.V. All rights reserved.

Table 1
A listing of the set of reservoir parameters used for modeling the Lorenz and Chua's circuit systems.

System	Reservoir parameter	Drive system	Response system
Lorenz	α	0.2	0.4
$\rho = 10$, b $= 8/3$	N	500	400
r $= 60$	T	2600	2800
Chua circuit	α	0.45	0.5
$\gamma = 10.0$, $\beta = 14.87$	N	500	550
a $= -1.27$, b $= -0.68$	T	4000	3800

2. Synchronization of reservoir computers

We begin with reservoir computers learning a chaotic system. Typically, its architecture contains three basic components: a linear input layer, a reservoir network with N nodes in the middle layer, and a linear output layer. Specifically, through an I/R (input to reservoir) coupler, an input vector $\mathbf{u}$ is fed into the reservoir network via the input weighted matrix $\mathbf{W}_{in}$, while an output vector $\mathbf{y}$ is coupled by an R/O (reservoir to output) coupler. Following Jaeger's design [6,15], the state vector $\mathbf{r}$ of the reservoir network is updated as follows:

$$\mathbf{r}(t+1) = (1-\theta)\mathbf{r}(t) + \theta \tanh\left(\mathbf{A}\mathbf{r}(t) + \mathbf{W}_{in}\begin{pmatrix} b_{in} \\ \mathbf{u}(t)\end{pmatrix}\right), \tag{1}$$

where $\mathbf{A}$ is the adjacency matrix of an Erdös–Rényi graph whose nonzero entries are drawn randomly from a uniform distribution [−1,1]. The parameter θ called a "leakage" rate lying in the range (0, 1) and $b_{in} = 1$ is the bias. The elements in $\mathbf{W}_{in}$ are chosen from a uniform distribution [−1,1]. Then, the output vector $\mathbf{y}$ of the reservoir computer is defined as

$$\mathbf{y}(t) = \mathbf{W}_{out}\begin{pmatrix} b_{out} \\ \mathbf{u}(t) \\ \mathbf{r}(t)\end{pmatrix}, \tag{2}$$

where b_{out} is the output bias and is typically set to 1. Here, $\mathbf{W}_{out}$ is the solely adjustable matrix needed to determined in the training stage.

Suppose that $\{\mathbf{s}(t)|t = 1, 2, \ldots, T\}$ is the training data for which we use $\mathbf{s}(t)$ to predict next forward $\mathbf{s}(t+1)$ based on Eqs. (1) and (2). In this context, the analytical expression of the output weights $\mathbf{W}_{out}$ is given by [15,16]

$$\mathbf{W}_{out} = \mathbf{Y}\mathbf{X}^{\top}(\mathbf{X}\mathbf{X}^{\top} + \lambda\mathbb{I})^{-1}, \tag{3}$$

where $\mathbb{I}$ is the identity matrix and λ (typically $\lambda = 1\times 10^{-8}$ a very small number) is the ridge regression parameter. Here, $\mathbf{X}$ (resp. $\mathbf{Y}$) is the matrix whose ith column is $[b_{out}; \mathbf{s}(i); \mathbf{r}(i)]$ (resp. $\mathbf{s}(i+1)$).

We first consider different reservoir computers for modeling the chaotic Lorenz system given by

$$dx/dt = \sigma(y - z), \tag{4}$$

$$dy/dt = -xz + rx - y, \tag{5}$$

$$dz/dt = xy - bz, \tag{6}$$

where $\sigma = 10$, $b = 8/3$, and $r = 60$. Utilizing the fourth order Runge–Kutta technique, we record 10 000 time points with time step $\Delta t = 0.02$. We use $\mathbf{u} = (x, y, z)$ as the input vector for training two reservoir computers with different reservoir parameters (details shown in Table 1). Clearly, the reservoir computers have capture the trajectory of the Lorenz system, where we show the x-variable as an example, see Fig. 1(a) and (b). After training stage, the first reservoir computer runs autonomously based on Eqs. (1) and (2) and generate a set of data points (x', y', z'). Note that the respective reservoir systems are called the drive system and the response system for convenience. Then, we use the y'-variable as the transmitted signal for which the input vector fed into the response system becomes $\mathbf{u} = (x'', y', z'')$. The initial values of x''_0 and z''_0 are chosen randomly. Interestingly, we find that by sending the y' signal, the response system will synchronize with the drive system. Specifically, we show rapid convergence of the response system to the drive system for x' and z' components in Fig. 1(c) and (d). Our findings point out that synchronization of the trained reservoir computers can be achieved by sending a common signal.

We further investigate this synchronization behavior in the Chua's circuit system. These circuit equations can be written as [17]

$$dx/dt = \gamma(y - x - f(x)), \tag{7}$$

$$dy/dt = x - y + z, \tag{8}$$

$$dz/dt = -\beta y, \tag{9}$$

where $f(x) = bx + 1/2(a-b)(|x+1| - |x-1|)$. Here, we select the parameters $(\gamma, \beta, a, b) = (10, 14.87, -1.27, -0.68)$ to be in the chaotic regime and compute 1×10^4 data points via the fourth order Runge–Kutta technique. Following the same

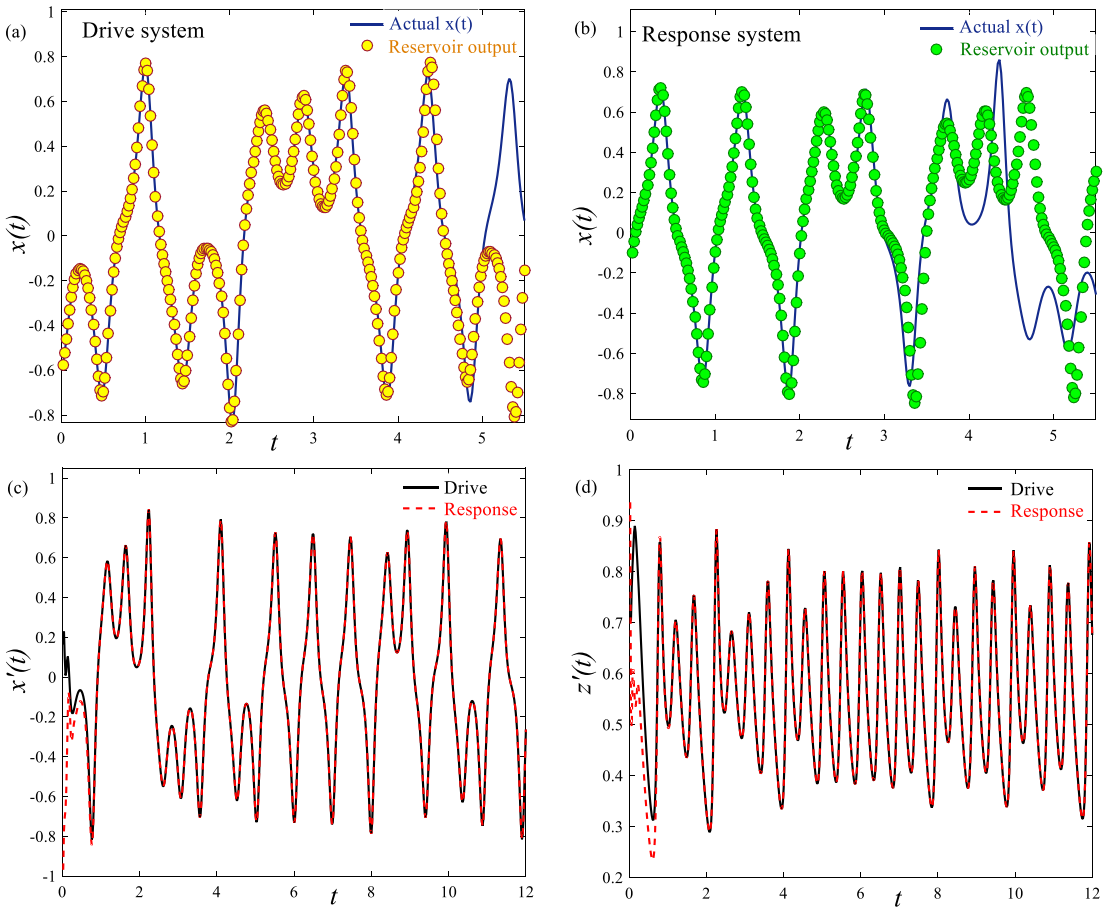

Fig. 1. Prediction output of the trained reservoir computer overlaid with actual data from the Lorenz system for (a) the drive system and (b) the response system. Complete synchronization of the drive system and the response system in term of (c) x'-variable and (d) z'-variable.

procedure as before, we train two different reservoir computers (details shown in Table 1) and build a drive–response configuration using x'-variable as the transmit signal. In this context, we show that the variable of the response system rapidly approaches that of the drive system (see Fig. 2(a)). This effect is further supported by observing the associated differences $\Delta y = |y' - y''|$ and $\Delta z = |z' - z''|$, which converge toward zeros, as described in Fig. 2(b). These findings confirm that synchronization is a common behavior in trained reservoir computers on learning a same chaotic system.

Moreover, since noise is ubiquitous in both nature systems and artificial systems, synchronization of trained reservoir computers is unavoidably subject to observational noise. Here, we investigate the effect of observational noise on this synchronization phenomenon and take the previous selected Lorenz system as an example. We train the reservoir computers in the same manner as previous. In the synchronization stage, we introduce white noise into the y' transmitted signal with the different levels. Fig. 3(a) shows the variable x'' of the response system versus its drive counterpart x' with respect to three different levels (i.e., signal-to-noise ratio (SNR)). The result hints that this synchronization behavior can tolerate high level noise. We further investigate how the level of noise (i.e., SNR) influences the ability of synchronization and introduce the average differences $\langle \Delta x \rangle = \frac{1}{n}\sum_{i=1}^{n} \Delta x_i$ and $\langle \Delta z \rangle = \frac{1}{n}\sum_{i=1}^{n} \Delta z_i$ to quantify the performance. We find that with the increase of the SNR, the average differences $\langle \Delta x \rangle$ and $\langle \Delta z \rangle$ decrease rapidly, see Fig. 3(b). Results demonstrate that synchronization of different reservoir computers is robust even when contaminated with strong observational noise.

3. The cascaded synchronization scheme for recovering the masking signal

Finally, as a potential application, we show that synchronization of trained reservoir computers can be used for chaotic signal masking and recovery. In signal masking, a speech masking signal $s(t)$ is added at the transmitted signal x'. Since synchronization of trained reservoir computers is found to be robust in the above section, the combined noised signal $m(t) = s(t) + x'$ can be adopted as the driving signal. By constructing a cascaded synchronization scheme [18,19], the masking signal is regenerated at the receiver. Specifically, we use the $m(t)$ as a common signal to link the drive system and

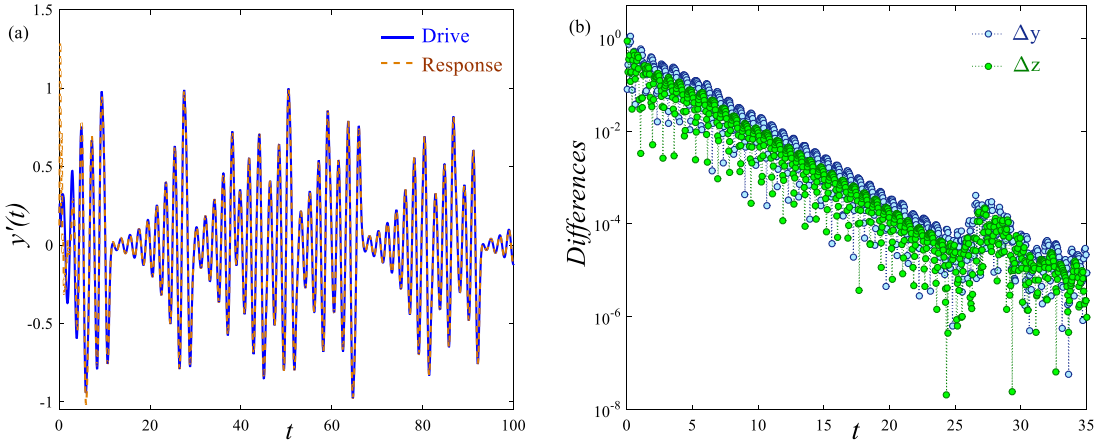

Fig. 2. Synchronization of the driven system and the response system with the x'-driven: (a) Solid line (dashed line) represents y' (y'') in function of time. (b) The differences $\Delta y = |y' - y''|$ and $\Delta z = |z' - z''|$ as a function of time. Trajectories of the (y', z') subsystems from (c) the drive system and (d) the response system. The initial transient states are shown in the orange line (c) and the red line (d), respectively.

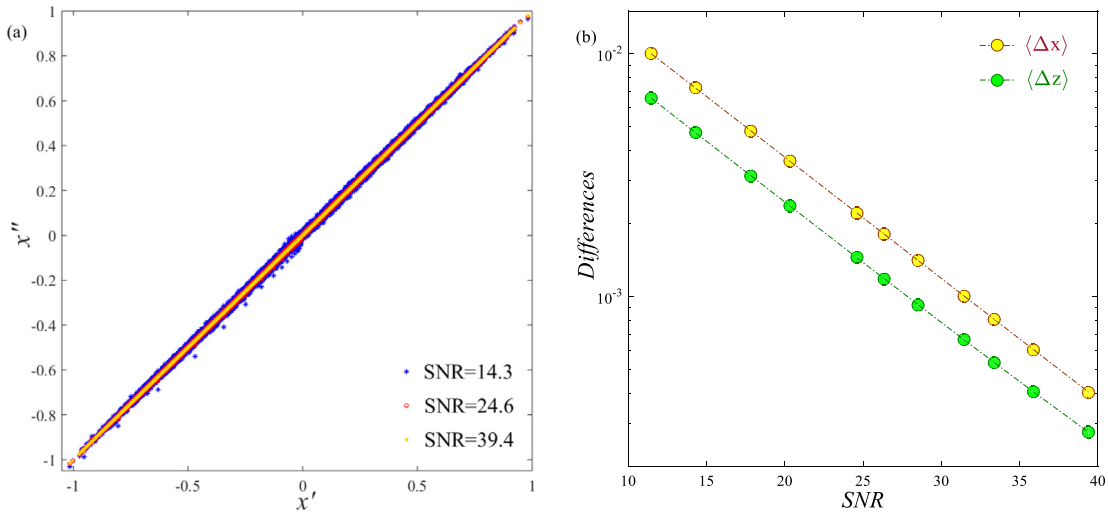

Fig. 3. Noise effect on synchronization of trained reservoir computers in learning the Lorenz system: (a) The response variable x'' vs. its drive counterpart x' in case of different levels of noise. (b) The average differences $\langle \Delta x \rangle$ and $\langle \Delta z \rangle$ as a function of SNR.

the response-1 system, while from response-1 system to the response-2 system, the y'' is the driving signal, as illustrated in Fig. 4. We show that our cascaded synchronization scheme can reproduce the masking signal and take the Lorenz system as an example. Here, the masking signal $s(t)$ is a segment of speech. Results show that the x'''-variable of the response-2 rapidly approaches to the x'-variable of the drive system as time progresses, see Fig. 5(a). In this context, the masking signal (i.e., speech waveform) is recovered by computing $s'(t) = m(t) - x'''$, which is comparable to the original one, see Fig. 5(b) and (c). Our findings reveal that synchronization behavior of reservoir computers can be used for reproducing the masking signal.

4. Conclusion

We investigate the modeling of chaotic systems via reservoir computers with different reservoir parameters. We find that by sending a common signal, synchronization will be achieved among these trained reservoir computers. We further show that this synchronization behavior is robust even the transmitted signal is contaminated by observational noise with high level. Remarkably, by constructing a cascaded synchronization scheme, we demonstrate that it can be used for recovering a chaotic masking signal. Our findings reveal that synchronization is a universal feature in reservoir computers upon learning a same chaotic system. Note that here we only investigate complete synchronization. A more intriguing open problem is whether or not other types of synchronization, such as lag synchronization [20], generalized synchronization [21], and phase synchronization [22], can also be achieved in reservoir computers.

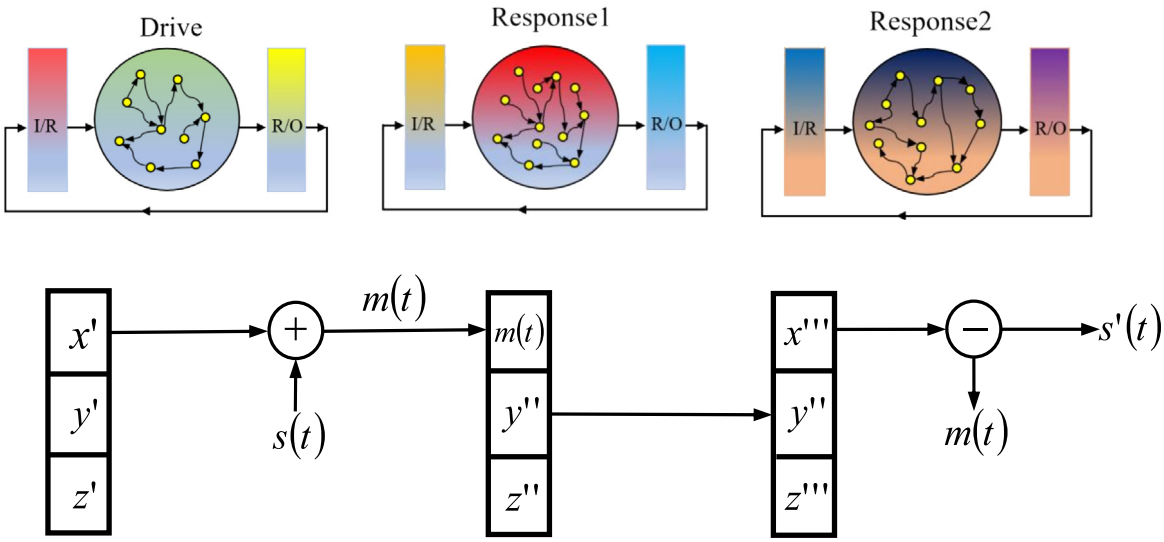

Fig. 4. Schematic illustration of chaotic signal masking system.

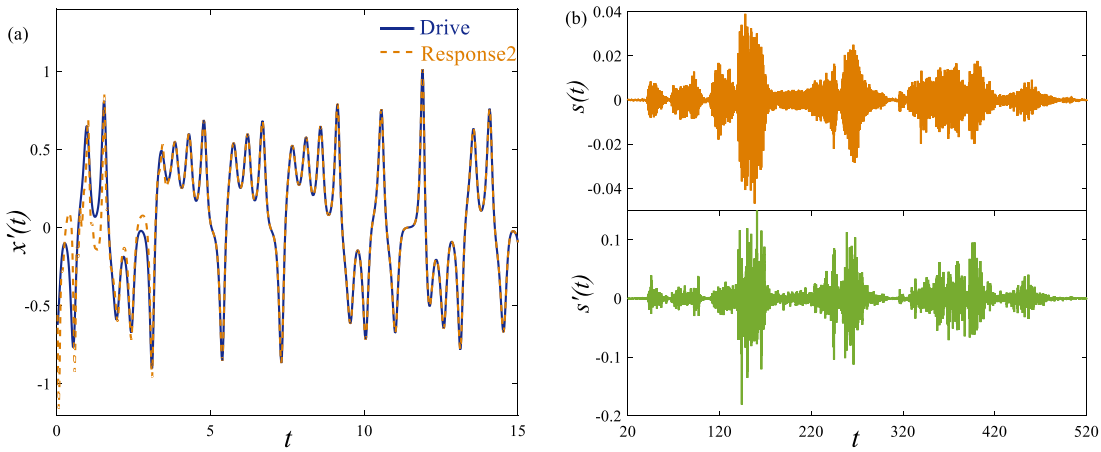

Fig. 5. (a) Synchronization of trained reservoir computers. Speech waveforms of (b) original and (c) recovered.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (NNSFC) under Grant Nos. 11875042 (C. Gu), 11505114 (C. Gu), 11805128 (T. Weng), and 10975099 (H. Yang); the Program for Professor of Special Appointment (Orientational Scholar) at the Shanghai Institutions of Higher Learning under Grant Nos. D-USST02 (H. Yang) and QD2015016 (C. Gu); and the Shanghai project for construction of top disciplines under Grant No. USST-SYS-01 (H. Yang, C. Gu, and T. Weng).

References

[1] D. Silver, J. Schrittwieser, K. Simonyan, I. Antonoglou, A. Huang, Nature 550 (2017) 354–359.
[2] A. Mohammed, R. Minhas, J. Kurths, Q.M. Wu, M.A. Sid-Ahmed, Pattern Recognit. 44 (2011) 2588–2597.
[3] Y. LeCun, Y. Bengio, G. Hinton, Nature 521 (2015) 436–444.
[4] C.W. Hsu, C.J. Lin, IEEE Trans. Neural Netw. 13 (2002) 415–425.
[5] C.C. Lee, E. Mower, C. Busso, S. Lee, S. Narayanan, Speech Commun. 53 (2011) 1162–1171.
[6] H. Jaeger, H. Haas, Science 304 (2004) 78–80.
[7] J. Pathak, B. Hunt, M. Girvan, Z. Lu, E. Ott, Phys. Rev. Lett. 120 (2018) 024102.
[8] J. Pathak, Z. Lu, B. Hunt, M. Girvan, E. Ott, Chaos 27 (2017) 121102.
[9] T.F. Weng, H.J. Yang, C.G. Gu, J. Zhang, M. Small, Phys. Rev. E 99 (2019) 042203.
[10] P. Antonik, M. Gulina, J. Pauwels, S. Massar, Phys. Rev. E 98 (2018) 012215.
[11] Z.X. Lu, J. Pathak, B. Hunt, M. Girvan, R. Brockett, E. Ott, Chaos 27 (2017) 041102.

[12] Z.X. Lu, B. Hunt, E. Ott, Chaos 28 (2018) 061104.
[13] A. Arenas, A. Díaz-Guilera, J. Kurths, Y. Moreno, C.S. Zhou, Phys. Rep. 469 (2008) 93–153.
[14] L.M. Pecora, T.L. Carroll, Chaos 25 (2015) 097611.
[15] M. Lukoševičius, H. Jaeger, Comput. Sci. Rev. 3 (2009) 127–149.
[16] R.S. Zimmermann, U. Parlitz, Chaos 28 (2018) 043118.
[17] L.O. Chua, L. Kocarev, K. Eckert, M. Itoh, Int. J. Bifurcation Chaos 2 (1992) 705–708.
[18] L.M. Pecora, T.L. Carroll, Physica D 67 (1993) 126–140.
[19] K.M. Cuomo, A.V. Oppenheim, S.H. Strogatz, IEEE Trans. Circuits Syst. II 40 (1993) 626–633.
[20] S. Taherion, Y.C. Lai, Phys. Rev. E 59 (1998) 6247–6250.
[21] S. Boccaletti, J. Kurths, G. Osipov, D.L. Valladares, C.S. Zhou, Phys. Rep. 366 (2002) 1–101.
[22] E.R. Rosa, M.H. Hess, Phys. Rev. Lett. 80 (1997) 1642.